

The empirical recurrence for OEIS A251469: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A251469 records “Empirical recurrence of order 45”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 216$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 45, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 45 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251469 is “Number of $(n+1) \times (2+1)$ 0..5 arrays with every 2×2 subblock summing to a prime.”. It has offset 1 and begins

5896, 167554, 4789458, 137223466, 3932396176, 112715450380, 3230677312514, 92604214969034, ...

The entry states:

Empirical recurrence of order 45 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:24:00, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 216$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 216$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 432$ exact terms of the model returns order 45 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{45} c_i a(n-i),$$

c_1	41	c_{13}	8637637539734420	c_{25}	−7974966053985984329079168	c_{37}	−1750
c_2	1054	c_{14}	−139910208986057894	c_{26}	−20544159658413755386265664	c_{38}	9273
c_3	−59105	c_{15}	−298430999518499143	c_{27}	88894732145049297971453568	c_{39}	8291
c_4	−183871	c_{16}	7990267032020741713	c_{28}	115014037994331406236856320	c_{40}	−416
c_5	31070076	c_{17}	1958452066854399215	c_{29}	−655297402368294738812292096	c_{41}	−162
c_6	−107056232	c_{18}	−317034067051659428992	c_{30}	−340589391297815734534969344	c_{42}	553
c_7	−8502540303	c_{19}	327032320337975193798	c_{31}	3150856829740329670110806016	c_{43}	141
c_8	57803958218	c_{20}	8819262064447134500652	c_{32}	110762516821436893146710016	c_{44}	−21
c_9	1377355175580	c_{21}	−17397525422748358409932	c_{33}	−9585764171027910303645892608	c_{45}	−45
c_{10}	−12810017103288	c_{22}	−171538453752832737783576	c_{34}	2574168282062062639888859136		
c_{11}	−138703908331999	c_{23}	470358385745411409613688	c_{35}	17495830277063615756930383872		
c_{12}	1662861205752937	c_{24}	2298035081464283442680304	c_{36}	−7847072846053159094452224000		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 45, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $45 + 4$ terms removed returns order 45 as well, so $\deg q_{\min} = 45 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $45 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 45 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251469.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.